

Business Problem

In today's competitive retail environment, understanding customer behaviour is essential for driving business growth, improving customer satisfaction, and maximizing profitability. However, many organizations struggle to effectively utilize the large volumes of customer and sales data they collect.

The company in this project faces challenges in identifying key purchasing patterns, understanding customer preferences, and evaluating the performance of different product categories and customer segments. Without a centralized analytical system, decision-makers have limited visibility into critical business metrics such as customer spending habits, subscription engagement, product demand, and customer satisfaction levels. This lack of insight makes it difficult to develop targeted marketing strategies, improve customer retention, optimize product offerings, and allocate resources efficiently.

The company also lacks a clear understanding of how demographic factors such as age and gender influence purchasing behaviour, which limits its ability to personalize services and promotions. This project addresses these challenges by analyzing customer behaviour and building a comprehensive dashboard to help the company improve operational efficiency, increase revenue, enhance customer loyalty, and make smarter business decisions based on real-time insights.

Deliverables

1. **Data Preparation & Modeling (Python):** Clean and transform the raw dataset for analysis.
2. **Data Analysis (SQL):** Organize the data into a structured format, simulate business transactions, and run queries to extract insights on customer segments, loyalty, and purchase drivers.
3. **Visualization & Insights (Power BI):** Build an interactive dashboard that highlights key patterns and trends, enabling stakeholders to make data-driven decisions.
4. **Report and Presentation:** Write a clear project report summarizing findings and business recommendations. prepared a presentation that visually communicates insights and actionable recommendations to stakeholders.
5. **GitHub Repository:** Include all python scripts, SQL queries, and dashboard files in a well-structured repository.